Impulse changes momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect force over time to momentum change.

SUCCESS CRITERIA - I can...

1. Calculate impulse from force and duration.
2. Match impulse direction to momentum change.
3. Use impulse to find a final velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impulse” and “average” to explain your answer. Sentence starters are optional.

impulse: net force acting over time; change in momentum

average: one value representing an interval or a set of values

momentum: mass times velocity, including direction

01 Warm-up: bring back an earlier idea

An ideal circular orbit has GM=**225 m³/s²** and r=**9 m**. **Find speed**, inward acceleration and period; **explain how velocity changes**.

Word help: ideal = using stated simplifying conditions

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

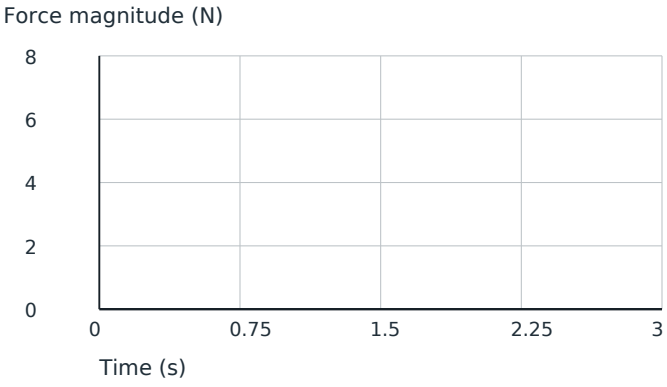
02 Worked example: explain the prediction

Reference: $J = F_{avg}\,\Delta t = \Delta p = p_f - p_i$.
J is impulse; F_{avg} is average net force. Signed force-time area gives impulse; i = initial, f = final.

Worked reasoning: A constant **6 N** force acts right for **2 s**. Impulse is **12 N·s** right, so momentum increases by **12 kg·m/s**.

Your turn: 1. **Draw the force-time rectangle** and **shade its area**.
2. **Explain why the two impulse** units are equivalent.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

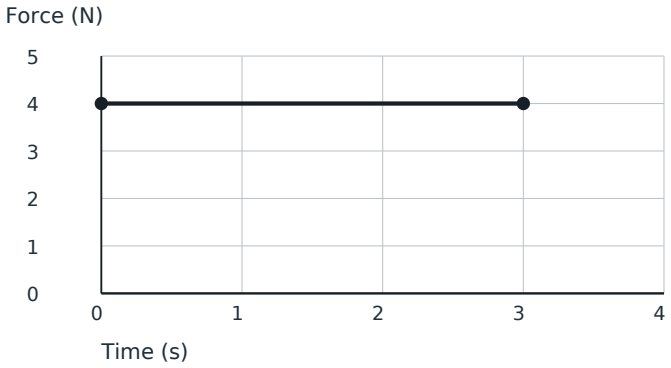


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Read the rectangular force-time graph**.
2. **Find the area** and **state what it means**.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

04 Apply the physics

A **0.50 kg** ball hits a wall while moving at **+8 m/s** and rebounds at **-6 m/s**. Contact lasts **0.020 s**.
1. **Label the before-and-after diagram**.
2. **Calculate impulse** and average net force on the ball.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	**After**

Label mass and velocity.

Marzano check - circle: ① ② ③

Impulse changes momentum

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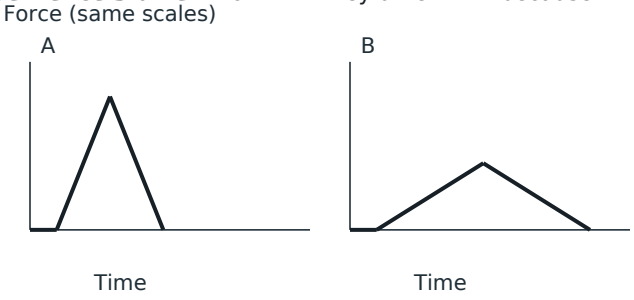
Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Two force pulses have equal signed area but different peak heights. What must be equal? What need not be equal?

Sentence starter: Both … . They differ in … because … .



Same scales: compare impulse and peak force.

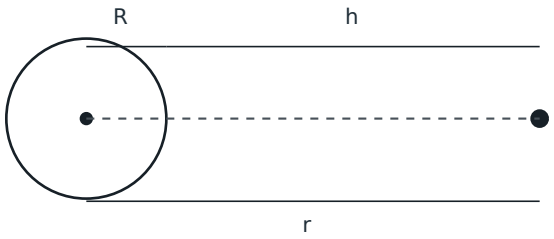
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

At what altitude h/R is a spherical planet's field **1/16** of its surface value? Repeat for **1/36**.
Label the masses, field or force relevant to the question.

Word help: altitude = height above a reference surface

Sentence starter: My design includes … . It does what is needed by … .

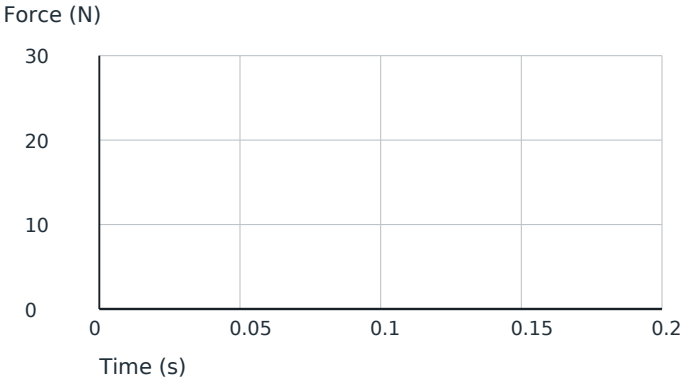


Marzano check - circle: ① ② ③

07 Exit check

A triangular force pulse has base **0.20 s** and peak **30 N**. **Find impulse** and velocity change for a **0.60 kg** object.

Sentence starter: I found … [unit]. This value means … in this situation.

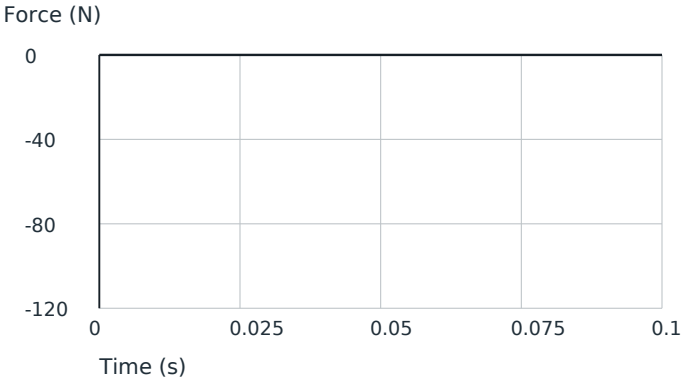


Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **2 kg** cart changes velocity from **+3** to **-2 m/s** in **0.1 s**.
1. **Find impulse** and average force.
2. **Sketch a force-time rectangle** with the correct signed area.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③